



Automated TV Defect Detection Using Generative AI

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Motivation & Use Case



Defect Detection in TV Manufacturing

- Ensuring product quality on the assembly line.
- Preventing faulty products from reaching customers.



Why This Problem Matters

- Reduces return and repair costs.
- Minimizes manual inspection expenses.
- Enhances customer satisfaction.

Current methods include manual checks or classical algorithms, which are often inefficient or prone to human error.

The Task: Automated Object Detection



Problem Statement:

Develop an automated system to detect and classify visual defects on TV screens.

Inputs:

Image of a TV screen from the production line.

Outputs:

Classification of defect type: Bezel Cracks, Corner Chips, Air Bubbles, etc.

Location of defect with a **bounding box**.

Synthetic Data Strategy: Leveraging Generative AI

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Models & Techniques

- **Stable Diffusion** with **ControlNet** and **Inpainting** to inject defects into images of flawless screens.
- **Object Detection Model** for defect identification.

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Variations

- **Camera angles** and lighting conditions.
- **Various defect types** (Bezel Cracks, Corner Chips, Air Bubbles), locations and light source.

Prompt Example

"Photorealistic spiderweb crack on black LCD TV screen, shattered glass texture, sharp jagged edges, realistic reflection, 4k resolution, industrial inspection photo style."

Evaluation Metrics: Ensuring Accuracy and Reliability



mAP (Mean Average Precision)

Measures **accuracy in defect localization** and classification.

Ensures precise identification of defect type and position.



Recall

Guarantees **detection of all actual defects**.

Prioritizes avoiding **false negatives** (missed defects) over false positives.